why the surface of the beach ball, like the surface of the garden hose discussed in preceding chapters, is two-dimensional.) They then considered shrinking the sphere until it is pinched down to a single point, as we illustrate with the sequence of shapes in Figure 11.3. This figure, and subsequent ones in this chapter, have been simplified by focusing in on the most relevant "piece" of the Calabi-Yau shape, but in the back of your mind you should note that these shape transformations are occuring within a somewhat larger Calabi-Yau space, as in Figure 11.2. And finally, Tian and Yau imagined slightly tearing the Calabi-Yau space at the pinch (Figure 11.4(a)), opening it up and gluing in another beach ball-like shape (Figure 11.4(b)), which they could then reinflate to a nice plump form (Figures 11.4(c) and 11.4(d)).

Mathematicians call this sequence of manipulations a flop-transition. It's as if the original beach ball shape is "flopped" over into a new orientation within the overall Calabi-Yau shape. Yau, Tian, and others noted that under certain circumstances, the new Calabi-Yau shape produced by a flop, as in Figure 11.4(d), is topologically distinct from the initial Calabi-Yau shape in Figure 11.3(a). This is a fancy way of saying that there is absolutely no way to deform the initial Calabi-Yau space in Figure 11.3(a) into the final Calabi-Yau space shown in Figure 11.4(d) without tearing the fabric of the Calabi-Yau space at some intermediate stage.

From a mathematical standpoint, this procedure of Yau and Tian is of interest because it provides a way to produce new Calabi-Yau spaces from ones that are known. But its real potential lies in the realm of physics, where it raises a tantalizing question: Could it be that, beyond its being an abstract mathematical procedure, the sequence displayed from Figure 11.3(a) through Figure 11.4(d) might actually occur in nature? Might it be that, contrary to Einstein's expectations, the fabric of space can tear apart and subsequently be repaired in the manner described?

The Mirror Perspective

For a couple of years after their 1987 observation, Yau would, every so often, encourage me to think about the possible physical incarnation of these flop transitions. I didn't. To me it seemed that flop transitions were merely a piece of abstract mathematics without any